Pangasinan State University

Urdaneta Campus
Bachelor of Science in Information Technology

Final Project

Mobile Application Development 1

Letter Quest

BSIT – 3A

Major in Web and Mobile Technologies

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Application Name: Letter Quest

Rationale

Young children often lose interest in traditional alphabet lessons because they feel repetitive and plain. When letters are presented as isolated, abstract shapes, kids quickly experience cognitive fatigue, making early learning feel like a chore rather than an adventure.

Letter Quest solves this by transforming literacy into a colorful, interactive visual exploration. By pairing vibrant letter cards with familiar objects, children can easily recognize and relate letters to the real world in a fun, stress-free format.

Features

1. **List of letters A to Z** – Bright cards for each letter showing the letter itself, a word, and a matching picture to help kids recognize and remember letters easily.
2. **Simple ABC Exploration** – Children can browse through all the letters and explore them at their own pace, making learning playful and stress-free.
3. **Shuffle Card Gallery** – A fun way to see different pictures with labels in random order, helping kids practice reading and word recognition naturally.
4. **4-Pics-1-Letter Game** – A short, engaging quiz where children pick the picture that starts with the displayed letter, reinforcing visual recognition and letter-to-object association.
5. **Favorites** – Kids can save their favorite letters or activities to revisit anytime and enjoy what they love most.
6. **Letter Sorting** – Letters can be sorted alphabetically or randomly to provide variety and keep learning interesting.
7. **Offline Progress Saving** – Allows children to view their saved favorite cards even without an internet connection, ensuring the most important learning materials are always available.

Technical Requirements

1. Hardware & Emulation Requirements

- **Emulator:** Medium Phone (e.g., Pixel 6 or Pixel 9) using the **Android-x64** system image.
- **API Level: API 36.0 (Android 16.0 "Baklava")**.

- **Memory (RAM):** 2 GB minimum (4 GB allocated for emulator performance).
- **Storage:** 150 MB free space (for APK installation and SQLite database persistence).

2. Software Requirements

- **Operating System:** Android 16.0 or higher.
- **Permissions:** Internal Storage read/write access (automatically handled by `sqflite` for saving favorite letters).
- **Audio Support:** None (intentional design choice to focus on visual discrimination).

3. Development Tools & Dependencies

- **Framework:** Flutter SDK.
- **Database:** `sqflite: ^latest` (for managing the Favorites collection).
- **State Management:** `shared_preferences: ^latest` (for user settings).
- **UI/Animation:** `animated_text_kit: ^latest` (for engaging letter transitions).

4. Installation Requirements

- **Package Format:** Release APK compiled for x86_64 (emulator) or ARM64 (physical device).
- **Sideloading:** Requires "Allow installation from unknown sources" to be enabled.
- **Persistence:** Local database initializes on first boot to ensure 100% offline functionality for all features.